

The one-dimensional second-order theorem

Claude, with Daniel Murfet

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Abstract

For $Z(N) = \int_0^b u^h \eta(u) e^{-\beta(Nu^k)^2 + \beta N u^k \xi(u)} du$ with ξ, η admitting first-order Taylor expansions at 0 with quadratic remainders, the density-transfer theorem (unit 83) applied to the exact density $\rho(r, s) = k^{-1} r^{p-1} \eta(r^{1/k}) e^{\beta s \xi(r^{1/k})}$ and the kernel Taylor bound (unit 84) give, for $N \geq 1$,

$$|Z(N) - C_0 N^{-p} - C_1 N^{-q}| \leq K' N^{-(h+3)/k},$$

$$C_0 = \frac{1}{k} \eta_0 S_p(a), \quad C_1 = \frac{1}{k} (\eta_1 S_q(a) + \beta \eta_0 a_1 S_{q+1}(a)),$$

with $p = (h+1)/k$, $q = (h+2)/k$, $a = \xi(0)$, $\eta_0 = \eta(0)$, $\eta_1 = \eta'(0)$, $a_1 = \xi'(0)$ and $S_\lambda(a) = \int_0^\infty s^{\lambda-1} e^{-\beta s^2 + \beta a s} ds$. This is the two-term case of `thm:TaylorTree` at $d = 1$ with the coefficient at the second candidate exponent identified. Zero `sorry/axiom`.

1 The things formalised

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noncomputable def oneDDensity ( p : ) ( k : ) ( : → ) ( r s : ) : :=
  1 / ( k : ) * r ^ ( p - 1 ) * ( ( r ^ (( k : ) ^ 1 )) * Real.exp ( * s * ( r ^ (( k : ) ^ 1 )) ) )
theorem oneDScale_eq_transferZ ... :
  oneDScale b N h k = transferZ (oneDDensity (((h : ) + 1) / k) k ) (b ^ k) N
theorem oneDDensity_expansion ... :
  DensityExpansion (oneDDensity p k ) ![p, q] ![0, 0]
  ![oneDCoeff a k, oneDCoeff a a k] (((h : ) + 3) / k) (b ^ k) 0
  (oneDEnvelope L C k)
theorem oneDScale_second_order ... :
  K' : , N : , 1 N →
  |oneDScale b N h k
  - (N ^ (-((h : ) + 1) / k))
    * (1 / (k : ) * ( * weightedMass a (((h : ) + 1) / k - 1)))
  + N ^ (-((h : ) + 2) / k))
    * (1 / (k : ) * ( * weightedMass a (((h : ) + 2) / k - 1)
    + * * a * weightedMass a (((h : ) + 2) / k))))|
  K' * N ^ (-((h : ) + 3) / k))

```

2 Proof strategy

The substitution $r = u^k$ (unit 44's `integral_Ioc_pow_mul_comp_pow` with $(u^k)^{1/k} = u$) writes the chart integral as a transfer integral with the exact density. The density expansion has two terms indexed by `Fin 2`: $r^{p-1} c_0(s)$ and $r^{q-1} c_1(s)$ with $c_0 = k^{-1} \eta_0 e^{\beta s a}$, $c_1 = k^{-1} e^{\beta s a} (\eta_1 + \beta s \eta_0 a_1)$, and remainder $r^{p+2/k-1} H(s)$, $H = C k^{-1} (1+s)^2 e^{3\beta L s}$, read off from the kernel Taylor bound at $u = r^{1/k}$. The weighted moments are finite because every coefficient is bounded by $e^{\beta s a'} (A + B s + D s^2)$, i.e. by sums of weighted quadratic kernels; the log moments at degree 0 are the fluctuation masses $S_\lambda(a)$ by `weightedMass`.

3 Roadblocks and resolutions

- The subtracted sum in the `rem` field is $A - (B + C)$ after `Fin.sum_univ_two`; state the algebraic identity in that shape.
- `rw [Fin.sum_univ_two]` rewrites only the first sum; use `simp only` to expand both sides before matching against the transferred bound.
- After `fin_cases` on `Fin 2` the vector entries are already reduced; the `Matrix.cons_val_*` `simp` arguments are then unused (linter).

4 Outcome

The first nonconstant- ξ, η coefficient beyond the leading term is formalised, in the explicit form predicted by Astra's consult ($C_1 = k^{-1}[\eta'(0)S_q(a) + \beta\eta(0)\xi'(0)S_{q+1}(a)]$). Next: the $d = 2$ equal-exponent anchored subtraction (the coefficient B seeing both axes) via the same transfer theorem, and the face-parameter version of the second-order theorem under the strict gap $q < \min_i p_i$.